

```

SelectedIndexChanged(object sender,
EventArgs e)
{
    serialPort1.PortName = PortCombo.
    Text;
    serialPort1.BaudRate = 115200
    buttonStart.Enabled = true;
    buttonStop.Enabled = false;
}

```

On pressing the start button, the serial port is enabled. The start button gets disabled and the stop button is enabled using following code:

```

private void buttonStart_
Click_1(object sender, EventArgs e)
{
    if (serialPort1.IsOpen == false)
    {
        serialPort1.Open();
        buttonStart.Enabled = false;
        buttonStop.Enabled = true;
    }
}

```

On pressing the stop button, the serial port is closed, start button is enabled and stop button is disabled.

```

private void buttonStop_
Click_1(object sender, EventArgs e)

```

```

{
    if (serialPort1.IsOpen)
    {
        serialPort1.Close();
        buttonStart.Enabled = true;
        buttonStop.Enabled = false;
    }
}

```

Now, to create an event of the track bar position from the incoming serial port data, select the serial port control in the form1 panel. Click on 'events' in the 'properties' tab, double click on 'serialPort1_DataReceived' event and add the following code:

```

private void serialPort1_
DataReceived(object sender, System.IO.
Ports.SerialDataReceivedEventArgs e)
{
    RxData = serialPort1.ReadByte();
    this.Invoke(new
    EventHandler(ChangeTrackbar));
}

```

As soon as the data is received, the source code above reads the existing byte from the serial port and takes the code to new event handler 'ChangeTrackbar' as defined below:

```

private void ChangeTrackbar(object
sender, EventArgs e)
{
    trackBarPot1.Value = RxData;
    //Set potentiometer input value
    (4.84)
    textBoxPot1.Text = Convert.
    ToString(RxData*4.84/255);
}

```

The handler sets the 'trackBarPot1' to the position as received from the serial port and stores in the variable RxData. The users may change the value of 4.84 to their own reference voltage. The voltage value is given as follows:

$ADC \text{ Value} \times V_{REF}/255$, for 8-bit data.

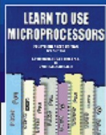
You can change the source code demonstrated in the article to your application, which may be an analogue centigrade sensor (e.g., LM 35) reading the temperature of a bearing or a room.

EFY note. The source codes of this article are available on www.efymag.com. ●

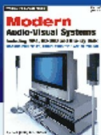
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